



Original Article

Correlates of insomnia among the adults during COVID19 pandemic: evidence from an online survey in India



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ABSTRACT

Background: Psychosocial distresses are often associated with sleep disorders. This is found as a common population level observation during natural calamities or emergency situations like a pandemic. The current study was conducted to determine the magnitude of self-reported insomnia, and to identify its associated factors among the apparently healthy Indian adult population having access to social media during the lockdown period of COVID 19 pandemic.

Methods: A cross-sectional, region-stratified survey was conducted from April 20 – May 19, 2020 in India using online questionnaire comprising Perceived Stress Scale, Generalized Anxiety Disorder Scale, Insomnia Severity Index, variables related to concern, fear and apprehension about COVID19 pandemic, and different socio-clinical variables. The questionnaire was disseminated via different social media platforms and responses from 1081 respondents were included in the analysis.

Results: Moderate clinical insomnia was present in 13.32% (95% confidence interval [CI]: 12.23%–14.50%), and severe clinical insomnia was present in 1.85% (95% CI: 1.08%–3.14%). For higher levels of insomnia mild anxiety was associated with a proportional odds ratio (OR) of 1.83 (95% CI: 1.24–2.71), and moderate and severe anxiety had an OR of 4.01 (95% CI: 2.57–6.25). Factors associated with higher severity of insomnia were middle age group (36–50 years), being a professional, living alone, perceiving COVID as serious disease, urban residence and those having comorbidity. The OR for being concerned for staying indoors was 5.09 (95% CI: 4.09–5.33), whereas the OR for fear of scarcity in availability of masks and sanitizers was 1.48 (95% CI: 1.26–1.75); and the OR for fear of unavailability of essential goods was 3.20 (95% CI: 2.06–4.98).

Conclusions: Psychosocial factors like generalized anxiety, higher age, higher perceived severity, and various concerns related to COVID19 pandemic were identified to be associated with higher degrees of insomnia. Necessary psychosocial interventions, may be required to address the burden of insomnia in a timely and long-term mode.

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1. Introduction

The burden of mental morbidities in general have been growing on a global scale. In India one in every seven person is reported to

be suffering from mental disorder(s) [1]. The spread of the novel coronavirus infection (COVID-19 or n-CoV-2019 or SARS-Cov-2) is causing widespread concern, fear and stress on a global scale, a reaction very natural to the uncertainties imposed by a pandemic situation [2,3]. The stern pandemic control actions like initiating pan-India lockdown, enforcement of strict social distancing policies have been extremely challenging putting the residents through psychosocial stress which is further complicated by a heightened sense of susceptibility and severity of the disease [4–7].

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Psychosocial distress is often associated with sleep disorders and is a common population level observation during natural calamities or emergency situation like a pandemic [8–10].

Insomnia is a disorder of sleep characterized by difficulty in at least initiation, maintenance or quality of sleep often associated with day-time consequences [11]. It is documented as one of the prevalent health issues among those who experienced quarantine or isolation [12]. During the time of COVID pandemic and also during the containment measures being enforced, researchers in several countries have extracted the correlates of insomnia, focusing primarily on psychological factors like generalized anxiety, increased stress and fear [13–18]. Age, gender along with education, occupation and family arrangements are documented to have significant impact on development of insomnia [14,19,20]. However, concern, apprehension and fear regarding different aspects of impact of the pandemic and its containment and/or mitigation measures, are conceptualized to affect sleep in a similar fashion like anxiety or stress. But, evidence regarding their roles is still lacking. However, in the context of insomnia in general, during the ongoing COVID19 pandemic, evidence is lacking from Indian population. Identification of modifiable and behavioral correlates of insomnia is important since the coupling of insomnia with psychosocial morbidity is a vicious one and require early sustainable intervention [21,22].

As the nation starts to ‘unlock’ the country and its economy, a healthy adult workforce is required. The problem of insomnia may just be the tip of the ice-berg of a much larger yet amenable problem. It is important from a policy perspective that physiological needs like sleep are sufficiently met. Lockdown and social distancing permitted the role of digital media to increase manifolds in terms of communication and source of information. It is thus rational also from feasibility perspective to survey the apparently healthy adults through online data collection technique, that will by virtue of the design account for the baseline effect of variable use of digital media. The current study was conducted to find out the prevalence of self-reported insomnia, and also to identify its associated factors among the Indian adult population having access to social media. In order to advocate for precision-based behavioral intervention and psychotherapeutic aid in the context of insomnia enumeration of the burden and its determinants appear cardinal.

2. Materials and methods

2.1. Procedures

2.1.1. Study type and population

An observational analytical online questionnaire-based survey with cross-sectional design was conducted among the social media users from India. The data collection for this study was conducted over one-month duration starting from April 20 – May 19, 2020. Clearance was obtained from the Institutional Ethics Committee of Medical College, Kolkata, West Bengal, India. Individuals who had access to social media platforms like Facebook®, Twitter®, Instagram®, LinkedIn®, were considered as the study population. Also, access to WhatsApp® and Email service(s) was considered necessary only for questionnaire dissemination among the selected participants. Adult population (18–65 years) and Indian by nationality who were currently living in India since the beginning of the nation-wide Lockdown on March 25, 2020 were included in the study. Those having critical illness or under palliative care were excluded. Participants diagnosed with any sleep disorder or cognitive or psychiatric illness or those reported to be on psychotropic or sedative medication were also excluded from this study through skip patterns incorporated in the online questionnaire.

Those who participated in the study provided an online informed consent before responding to the online questionnaire.

2.1.2. Selection of the participants

Based on the observations from a pilot study (proportion of subthreshold and clinical insomnia ~45%) the minimum required sample size was estimated to be 940, considering 5% precision and 90% power of the study with a design effect of 2. The response rate was taken as minimum 80% out of total distribution of the questionnaire. This yielded a target sample size of 1128, for which 188 responses were targeted from each zone. Social media platforms were searched by name of different states and union-territories as per the zones, where zones were used to stratify the sample. The resultant open-ended list was used to select desired number of participants in different zones through random sequences. The participants were contacted through their available contact information (email or WhatsApp number) and the Google form was shared. Finally, a total of 1081 responses were included in the final analysis with 211 from Eastern zone, 203 from Northern zone, 171 from Western zone, 174 from Southern zone, 151 from Central zone and 171 from North-eastern zone. Fig. 1 summarizes the zonal structure and selection of participants from each zone. The collected data was validated through re-test with the same questionnaire sent through email to randomly selected 10% respondents in each day. These re-test responses were used for validation purpose only and not included in the final sample for analysis.

2.1.3. Questionnaire

The pre-designed pre-tested online questionnaire (Google form), used for data collection was comprised of questions on Socio-demographic and clinical information, awareness on COVID pandemic and cases in the surroundings, sources of fear, apprehension and concern, perceived stress, levels of insomnia and

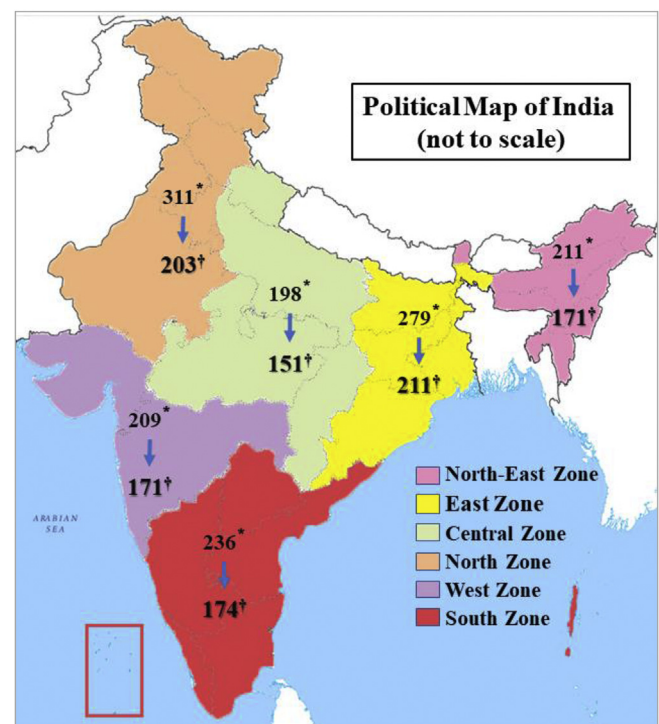


Fig. 1. Participants selected from the different zone in India. * indicates the number of participants in the respective zone who accessed the data collection form, † indicates who responded and submitted the form.

anxiety. The collected data using predesigned Google form (Google LLC, California, USA), was auto-entered into a linked pre-generated Google sheet (Google LLC, California, USA). The questionnaire was designed through brain-storming sessions with subject experts from the disciplines of psychiatry, clinical psychology and health education, and was subsequently validated. The participants were enquired about their age, gender, residence, current living arrangement, employment status, highest educational level, and sources of information regarding COVID pandemic. They were also enquired whether there were any confirmed and/or suspect cases in their immediate social surrounding and/or relatives and close friends. A single-item three-point Likert-type question determined participants' perception regarding disease severity.

2.2. Measurements

2.2.1. Insomnia

Insomnia Severity Index was used to measure self-reported insomnia among the participants [23]. The index consisted of seven-items on five-point Likert-type scale, each item scored from 0 to 4. The Cronbach's alpha coefficient for this scale was 0.89 in the current study. Based on the total scale score calculated by summing up item-specific scores, participants were classified into four groups viz. no clinically significant insomnia (0–7), subthreshold insomnia (8–14), moderate severe clinical insomnia (15–21) and severe clinical insomnia (22–28).

2.2.2. Generalized anxiety

Anxiety was measured by the seven-item Generalized Anxiety Disorder Scale (GAD-7) each item measured in a four-point Likert-type scale scored from 0 to 3 [24]. Based on the total summated score in GAD-7 the participants were grouped into four anxiety groups e.g. minimal or no anxiety (0–4), mild anxiety (5–9), moderate anxiety (10–14) and severe anxiety (15–21). The observed Cronbach's alpha coefficient was 0.86 according to the responses in the current study.

2.2.3. Perceived stress and areas of concern

Perceived stress was measured with the help of validated four item Perceived Stress Scale (PSS-4) [25], with each item measured in a five-point Likert-type scale having a score of minimum 0 and maximum 4 for each item. The total PSS-4 score was calculated by adding up all the item-specific scores. Higher level of perceived stress was denoted by a higher total score in PSS-4 scale. The Cronbach's alpha coefficient for PSS-4 in the study sample was 0.52, with substantial median responses. Upon item-analysis by rating scale model the discrimination was observed to be low with an acceptable difficulty level. The respondents were asked to select the applicable sources of fear, concern and apprehension from three separate lists prepared in consultation with mental health practitioners and clinical psychologists.

2.3. Statistical analysis

Statistical analysis was performed using STATA 14.0 software (StataCorp LLC, College Station, Texas, USA). Confidentiality was maintained while cleaning and storing the data for analysis. Continuous variables e.g. age in completed years, total PSS-4 score, total insomnia score and total GAD-7 score were found to be not normally distributed and hence were described with median and inter-quartile range (IQR). The proportions or prevalence of different factors were calculated with clustered robust standard error taking into account the clustering effect of the zones. The categorical variables in the study were tested for trends across different zones by chi-square test. The differences in median values

of the continuous study variables were tested by Kruskal–Wallis test. While analyzing different factors for their association with anxiety levels, the regression models were built combining responses from all the zones. Partially complete responses i.e. those with missing values were dropped from the regression models.

For the purpose of understanding the multi-variable relationship between different factors and the different levels of insomnia, moderate severe and severe insomnia were clubbed into 'clinical insomnia'. Similarly, for anxiety categories moderate and severe levels of anxiety were combined to ultimately yield three ordered categories of anxiety – no anxiety, mild anxiety and moderate to severe anxiety. A regression models was created considering age group, gender, marital status, education, employment, any diagnosed illness, perception of disease severity, whether living alone or not, level of self-reported anxiety and total PSS-4 score as the explanatory variables. Since, the outcome variable in this case was ordinal in nature and the proportional odds assumption was violated, generalized ordinal regression models were built allowing for partial fulfilment for proportional odds in some variables. The generalized ordinal model was in fact composite of two binary logistic regression models. In the model there were two components, the first one showing the probability of being in higher level of anxiety (mild to severe) compared to no anxiety, and the second part yields the probability of being in moderate to severe anxiety level tested against no to mild anxiety. A P-value of <0.05 will be considered statistically significant.

3. Results

3.1. Background analysis

Total 1081 responses were included in the analysis. The median age was 32 years with an inter-quartile range (IQR) of 15 years. Table 1 describes the socio-demographic characteristics of the participants. More than half of both male and female respondents were currently married and belonged to younger age group. At the time of the survey 4.31% of the students were reporting to their institutions while 14.75% of the employed were going outside of their home for their work. Among the participants, 19.33% had at least one diagnosed illness.

There was no statistically significant difference in responses from different zones and degrees of anxiety. The median score in GAD scale was 4 (IQR: 7). Overall, mild, moderate and severe levels of anxiety was reported by 27.94%, 16.00% and 2.68% respectively. Table 2 shows the proportions of different level of anxiety among the respondents.

3.2. Prevalence of insomnia

Table 2 also describes the proportion of different degrees of insomnia in different zones. Overall median score in ISI scale was 6 (IQR: 9). Among the respondents 13.32% (95% CI: 12.23%–14.50%) and 1.85% (95% CI: 1.08%–3.14%) had respectively moderate and severe degree of clinical insomnia.

3.3. Factors associated with insomnia

The ordinal regression model describing the association between different levels of insomnia and its selected predictors among the overall surveyed population is depicted in Table 3.

3.3.1. Association of clinico-social factors with insomnia

Overall, compared to the younger age group, middle age group (36–50) had an odds of 1.50 (95% CI: 1.02–2.21) of progressing to higher degree of insomnia. However, the odds among the middle

Table 1
Socio-demographic and clinical profile of the participants.

Socio-demographic and clinical factors	Zones						Total	P-value
	East	North	West	South	Central	North-East		
Age group	n = 211	n = 203	n = 171	n = 174	n = 151	n = 171	n = 1081	
18 – 35 years	145 (68.72)	143 (70.44)	108 (63.16)	120 (68.97)	104 (68.87)	118 (69.01)	738 (68.27)	0.978
36 – 50 years	52 (24.64)	48 (23.65)	51 (29.82)	41 (23.56)	37 (24.50)	42 (24.56)	271 (25.07)	
51 – 65 years	14 (6.64)	12 (5.91)	12 (7.02)	13 (7.47)	10 (6.62)	11 (6.43)	72 (6.66)	
Gender	n = 211	n = 203	n = 171	n = 174	n = 151	n = 171	n = 1081	
Female	93 (44.08)	81 (39.90)	76 (44.44)	64 (36.78)	65 (43.05)	72 (42.11)	451 (41.72)	0.678
Male	118 (55.92)	122 (60.10)	95 (55.56)	110 (63.22)	86 (56.95)	99 (57.89)	630 (58.28)	
Marital Status	n = 211	n = 203	n = 171	n = 174	n = 151	n = 171	n = 1081	
Married	102 (48.34)	107 (52.51)	95 (55.56)	100 (57.47)	75 (49.67)	93 (54.39)	572 (52.91)	0.481
Currently not married	109 (51.66)	96 (47.29)	76 (44.44)	74 (42.53)	76 (50.33)	78 (45.61)	509 (47.09)	
Residence	n = 211	n = 203	n = 171	n = 174	n = 151	n = 171	n = 1081	
Rural	22 (10.43)	12 (5.91)	16 (9.36)	9 (5.17)	11 (7.28)	12 (7.02)	82 (7.59)	0.353
Urban	189 (89.57)	191 (94.09)	155 (90.64)	165 (94.83)	140 (92.72)	159 (92.98)	999 (92.41)	
Highest Educational Level	n = 211	n = 203	n = 171	n = 174	n = 151	n = 171	n = 1081	
Up to completed Class XII	18 (8.53)	18 (8.87)	13 (7.60)	6 (3.45)	14 (9.27)	16 (9.36)	85 (7.86)	0.589
Non-Professional (Graduates and above)	86 (40.76)	81 (39.90)	70 (40.94)	64 (36.78)	59 (39.07)	70 (40.94)	430 (39.78)	
Professional	107 (50.71)	104 (51.23)	88 (51.46)	104 (59.77)	78 (51.66)	85 (49.71)	566 (52.36)	
Occupation	n = 211	n = 203	n = 171	n = 174	n = 151	n = 171	n = 1081	
Unemployed	30 (14.22)	23 (11.33)	25 (14.62)	23 (13.22)	15 (9.93)	21 (12.38)	137 (12.67)	0.578
Employed or Home-maker	131 (62.09)	133 (65.52)	117 (68.42)	122 (70.11)	100 (66.23)	109 (63.74)	712 (65.86)	
Student	50 (23.70)	47 (23.15)	29 (16.96)	29 (16.67)	36 (23.84)	41 (23.98)	232 (21.46)	
Living arrangement	n = 211	n = 203	n = 171	n = 174	n = 151	n = 171	n = 1081	
Living alone	20 (9.48)	12 (5.91)	14 (8.19)	13 (7.47)	8 (5.30)	11 (6.43)	78 (7.22)	0.643
Living with Parents/Parents In-law	119 (56.40)	119 (58.62)	87 (50.88)	92 (52.87)	90 (59.60)	97 (56.73)	604 (55.87)	0.562
Living with Grandparents/Grandparents In-law	9 (4.27)	9 (4.43)	8 (4.68)	6 (3.45)	7 (4.64)	9 (5.26)	48 (4.44)	0.981
Living with Spouse/Partner	88 (41.71)	94 (46.31)	87 (50.88)	85 (48.85)	68 (45.03)	82 (47.95)	504 (46.62)	0.560
Living with Children	49 (23.22)	53 (26.11)	56 (32.75)	47 (27.01)	38 (25.17)	47 (27.49)	290 (26.83)	0.446
Diagnosed illnesses^a	n = 42	n = 38	n = 30	n = 37	n = 28	n = 34	n = 209	
Cardiovascular illness	6 (2.84)	4 (1.97)	6 (3.51)	1 (0.57)	4 (2.65)	5 (2.92)	26 (2.41)	0.558
Respiratory illness	3 (1.42)	2 (0.99)	1 (0.58)	2 (1.15)	1 (0.66)	2 (1.17)	11 (1.17)	0.969
Metabolic disorders	22 (10.43)	21 (10.34)	15 (8.77)	26 (14.94)	16 (10.60)	18 (10.53)	118 (10.92)	0.568

Numbers within the parenthesis denote percentages within the responses in each zone. 'n' represents the number of responses analyzed in each zone for the different variables. P-values mentioned in the table are obtained by chi-square test for statistical difference.

^a Only those who reported to have any diagnosed illness.

Table 2
Anxiety and insomnia among the respondents from different zones.

	Zones						Total	P-value
	East	North	West	South	Central	North-East		
Levels of Anxiety	n = 211	n = 203	n = 171	n = 174	n = 151	n = 171	n = 1081	
Minimal or no anxiety	101 (47.87)	113 (55.67)	89 (52.05)	109 (62.64)	77 (50.99)	88 (51.46)	577 (53.38)	0.427
Mild anxiety	67 (31.75)	57 (28.08)	49 (28.65)	40 (28.65)	42 (27.81)	47 (27.49)	302 (27.94)	
Moderate anxiety	34 (16.11)	29 (14.29)	28 (16.37)	24 (13.79)	26 (17.22)	32 (18.71)	173 (16.00)	
Severe anxiety	9 (4.27)	4 (1.97)	5 (2.92)	1 (0.57)	6 (3.97)	4 (2.34)	29 (2.68)	
Degrees of Insomnia	n = 211	n = 203	n = 171	n = 174	n = 151	n = 171	n = 1081	
No clinically significant insomnia	120 (56.87)	120 (59.11)	106 (61.99)	104 (59.77)	91 (60.26)	98 (57.31)	639 (59.11)	0.976
Sub-threshold insomnia	55 (26.07)	54 (26.60)	40 (23.39)	42 (24.14)	40 (26.49)	47 (27.49)	278 (25.72)	
Moderate to severe clinical insomnia	29 (13.74)	26 (12.81)	21 (12.28)	26 (14.94)	18 (11.92)	24 (14.04)	144 (13.32)	
Severe clinical insomnia	7 (3.32)	3 (1.48)	4 (2.34)	2 (1.15)	2 (1.32)	2 (1.17)	20 (1.85)	

Numbers within the parenthesis denote percentages within the responses in each zone. 'n' represents the number of responses analyzed in each zone for the different variables. P-values mentioned are obtained by chi-square test for statistical difference.

age group respondents for progressing to clinical insomnia was 0.68 (95% CI: 0.46–0.99). Married respondent reported an OR of 0.44 (95% CI: 0.31–0.62) of getting clinical insomnia. However, for professionals an OR of 3.09 (95% CI: 1.16–8.24) was noted in favor of clinical insomnia. Those living alone had an odds of 3.01 (95% CI: 1.24–7.28) times for getting higher degree of insomnia. The OR for those having any diagnosed illness was 1.66 (95% CI: 1.27–2.17) for having higher degrees of insomnia. Among those not living alone older age (aOR: 0.30, 95% CI: 0.17–0.53) and being married (aOR: 0.63, 95% CI: 0.41–0.97) were found to be protective. Residents of urban area (aOR: 1.77, 95% CI: 1.00–3.13) and those having any diagnosed illness (aOR: 1.63, 95% CI: 1.14–2.33) were at risk.

3.3.2. Effects of awareness and perception regarding COVID19 on insomnia

In the overall model, Presence of any confirmed or suspect case of COVID19 in the surrounding was associated with some (sub-threshold or clinical) degree of insomnia (aOR: 1.29, 95% CI: 1.00–1.67), but was protective against development of clinical insomnia (aOR: 0.59, 95% CI: 0.43–0.81). The OR for higher grades of insomnia was almost thrice among those who perceived that there is a high chance of death from the disease. The odds in favor of clinical insomnia was 6.13 (95% CI: 2.94–12.77) among those who thought COVID illness was a moderate disease with some health complications. Among those who were not currently living alone, perceiving COVID as a severe disease was associated with an OR of

Table 3

Generalized ordinal regression model summarizing the effect of the factors associated with different degrees of insomnia among the respondents. (n = 1068).

Factors associated with different degrees of insomnia ^a	aOR (95%CI) for subthreshold and clinical insomnia	aOR (95%CI) for Progression to clinical insomnia
Age (Ref: 18–35 years)		
36–50 years	1.50 (1.02–2.21) ^d	0.68 (0.46–0.99) ^d
51–65 years	0.34 (0.21–0.57) ^d	0.81 (0.38–1.74)
Gender (Ref: Female)		
Male	0.76 (0.57–1.02)	1.26 (0.82–1.93)
Marital status (Ref: Not currently married)		
Currently married	0.94 (0.80–1.12)	0.44 (0.31–0.62) ^d
Residence (Ref: Rural) ^b		
Urban	1.62 (0.98–2.66)	1.62 (0.98–2.66)
Highest level of education (Ref: Up to completed class XII)		
Non-professional Graduates and above	0.75 (0.32–1.74)	1.44 (0.67–3.09)
Professional Graduates and above	0.46 (0.19–1.11)	3.09 (1.16–8.24) ^d
Occupation (Ref: Unemployed) ^b		
Student	1.12 (0.63–2.01)	1.12 (0.63–2.01)
Employed or homemaker	1.04 (0.55–1.96)	1.04 (0.55–1.96)
Living arrangement (Ref: Currently living with family and/or friends)		
Currently living alone	3.01 (1.24–7.28) ^d	2.59 (1.06–6.35) ^d
Any diagnosed non-COVID illness (Ref: No diagnosed illness) ^b	1.66 (1.27–2.17) ^d	1.66 (1.27–2.17) ^d
Any confirm &/or suspect cases in the surrounding &/or family and close friends (Ref: No such cases)	1.29 (1.00–1.67) ^d	0.59 (0.43–0.81)
Sources of COVID related information ^c		
Healthcare workers	1.53 (1.18–1.97) ^d	0.43 (0.31–0.60) ^d
Social media	1.48 (1.00–2.17) ^d	0.92 (0.72–1.18)
Perception on features of COVID (Ref: Mild disease with no complications)		
Moderate disease with limited health complications	2.68 (1.44–5.00) ^d	6.13 (2.94–12.77) ^d
Serious disease with high chance of death ^b	2.99 (1.59–5.63) ^d	2.99 (1.59–5.63) ^d
Generalized anxiety (Ref: No anxiety)		
Mild anxiety ^b	1.83 (1.24–2.71) ^d	1.83 (1.24–2.71) ^d
Moderate & severe anxiety	2.43 (1.57–3.75) ^d	4.01 (2.57–6.25) ^d
Perceived stress score ^b	1.00 (0.93–1.08)	1.00 (0.93–1.08)
Concerned over ^c		
Performing regular handwash with soap and water	0.97 (0.72–1.31)	3.09 (2.04–4.66) ^d
Dedicately avoiding touching of face	0.92 (0.69–1.22)	2.00 (1.19–3.36) ^d
Frequent use of sanitizers to keep hands clean ^b	1.14 (0.94–1.39)	1.14 (0.94–1.39)
Actively practicing covering face and mouth with bent elbow when coughing and sneezing	3.69 (2.17–6.28) ^d	0.34 (0.23–0.52) ^d
Regular use of masks to cover nose and mouth	0.58 (0.38–0.88) ^d	0.93 (0.60–1.43)
Practicing social distancing ^b	0.70 (0.51–0.97) ^d	0.70 (0.51–0.97) ^d
Staying indoors as part of lockdown ^b	5.09 (4.09–6.33) ^d	5.09 (4.09–6.33) ^d
Fearful about ^c		
Scarcity in availability of masks and sanitizers ^b	1.48 (1.26–1.75) ^d	1.48 (1.26–1.75) ^d
Non-availability of essential goods including medicines ^b	3.20 (2.06–4.98) ^d	3.20 (2.06–4.98) ^d
Apprehension of ^c		
Losing job of oneself or family-member ^b	0.87 (0.66–1.15)	0.87 (0.66–1.15)
Being infected ^b	1.21 (0.89–1.63)	1.21 (0.89–1.63)
Near and dear ones being infected	1.08 (0.80–1.47)	0.76 (0.45–1.26)

aOR: adjusted Odds Ratio, CI: Confidence Interval, Ref.: Reference category.

^a Fit-statistics for Model A: Pseudo R² is 23.58% log pseudolikelihood = -768.92, Model with partial proportional odds assumption was statistically fit (P = 0.859).^b Proportional odds assumption satisfied.^c The reference for each of these variable categories are their complementary categories.^d Statistically significant (at P-value <0.05) aOR.

4.08 (95% CI: 2.34–7.10) of having clinical insomnia. Healthcare workers or social media being the source of information, was associated with having some degree of insomnia but was found protective against clinical insomnia.

3.3.3. Effect of anxiety and stress on insomnia

Those having concern over regular use of soap had an OR of 3.09 (95% CI: 2.04–4.66), while those concerned over active efforts to avoid touching face had an OR of 2.00 (95% CI: 1.19–3.36). On the other hand, those concerned over active efforts to cover the mouth with bent elbow while coughing, were at risk of having some degree of insomnia (OR: 3.69, 95% CI: 2.17–6.28) but were protected from clinical insomnia (OR: 0.34, 95% CI: 0.23–0.52). Those having mild anxiety had a proportional OR of 1.83 (95% CI: 1.24–2.71) for higher levels of insomnia. An OR of

4.01 (95% CI: 2.57–6.25) was observed for respondents with moderate or severe anxiety.

3.4. Additional analysis

3.4.1. Information and perception regarding COVID19 pandemic

Among the respondents 30.53% reported to have received information about the COVID19 illness from healthcare workers. Among those receiving information from social media (48.10%), 69.14% perceived the disease to be of higher severity. Among those diagnosed with any illness 79.90% reported a perception of higher severity of COVID19 illness. Overall, 19.52% of the respondents reported to have confirmed or suspect case(s) of COVID19 in their surrounding or among their relatives. Among them 55.92% perceived the disease to be severe with high chance of death. On

the other hand, those who did not report any confirmed or suspect cases nearby or amongst relatives, 62.89% perceived a higher severity of the disease (Refer Table A1).

3.4.2. Perceived stress and different sources of fear, concern and apprehension

Overall, the median score for perceived stress as per PSS4 was 6 (IQR: 3). Fig. A.1 depicts the median scores of the PSS4 over different zones. The differences in scores in the different zones were statistically significant. The reported proportion of different factors related to the pandemic causing concern, fear or apprehension among the respondents are shown in Fig. A.2. The majority of the respondents (91.77%) reported to have stress regarding at least two issues related to the ongoing pandemic. The major cause of stress as reported by the respondents was the apprehension that near and dear ones might get infected (70.03%) followed by staying indoors (68.36%). The median score for perceived stress was statistically higher among those who reported concern over frequent handwashing with soap, avoiding touching face, frequent use of sanitizers, covering mouth while coughing and sneezing, regular use of mask, scarcity of personal protective materials, maintaining physical distance, staying indoors, and apprehension of loss of job.

4. Discussion

4.1. Key findings

Among the participants who were identified to have insomnia, majority had subthreshold insomnia. The risk factors for developing clinical insomnia were: being professionals, currently living alone, previously diagnosed with any morbidity, higher perceived severity of the COVID infection, afraid of not getting essential supplies and personal protective materials, and concerned over regular hand washing practices and staying indoors. Higher level of self-reported anxiety was also associated with clinical insomnia, while presence of generalized anxiety was a prevalent finding among the participants. Respondents belonging to middle age group were at risk of developing some degree of insomnia but were protected from developing clinical insomnia. Having any confirmed or suspect case in the surrounding, receiving information from healthcare workers and concerned over practice of cough hygiene had a likewise effect. However, social media as a source of COVID related information was associated with some degree of insomnia.

4.2. What is already known and what this study adds

Several studies conducted on insomnia among users of social media described a high burden focusing primarily the young population [26–28]. In the current study social media use was not investigated but was used as a mean to access the study population which was the eligible adult population. Considering age-variability, geographical variability and also adjusted for variable use of social media accounted for in the design conduct and analysis of the study, the reported prevalence of different grades of insomnia was higher than what is reported in India before the outbreak of COVID [29,30]. Roy et al. (2020) in their study during early phases of lockdown estimated sleep disturbance in a proportion of 12.5% participants which is considerably low comparing to current observations. It is thus safe to presume that with progression into lockdown and gradual expansive impact of the pandemic, insomnia has emerged as a major health concern.

Similar to the current findings the prevalence of insomnia among Chinese, Italian and Greek population during the pandemic and lockdown was around 20–40% [3,13,14,16,18,19]. Generalized anxiety disorder was also a common finding during the pandemic

in other Indian studies and also studies from abroad showing a proportion of around 10–35% [3,13,14,19,22,31–33]. However, the current findings of generalized anxiety was much higher in proportion. These proportions were comparable somewhat with those from European or Chinese studies probably due to the fact that the stage of the pandemic and lockdown in India while conducting this survey was comparable to those nations.

Though many researchers from around the globe have enquired into mental health of the population and also sleep disturbances, but studies explaining the correlates of sleep disturbances are scarce. Studies among Italian and Chinese population during the expansion of the pandemic found out that anxiety was a key correlate of insomnia [3,14]. The findings were supported by the current evidence. While some studies noted increasing age to be a risk factor for insomnia the current findings were not in support [13]. Probably night-time engagements of the younger age group in India by means of digital media, put them at an increased risk of insomnia.

COVID-related literatures reveal that knowledge about location of cases of COVID, spending more time for searching COVID related news, being a healthcare worker were some important factors associated with poor sleep [3,13,18]. It was noted that attaining a professional degree was associated with development of clinical insomnia. In the current study awareness about cases in the surrounding, several sources of fear, apprehension and concern regarding COVID situation yielded higher levels of insomnia. Recent literatures are in consonance to the finding of a grave perception of COVID severity to be associated with sleep difficulties [9,13]. The poor sleep among those diagnosed with other illnesses were conceptually consistent and require further study in the specific disease processes, which is beyond the scope of this article. Gualano et al. (2020) though did not investigate living arrangement as a predictor but as a proxy found out that those married and currently cohabiting were protected from sleep disturbance [14]. The current results show that those living alone at the time of the study were at risk of poor sleep indicating a possible role of loneliness. While literature on sleep disturbance during COVID outbreak reported that female gender was a risk factor for insomnia, the current article did not demonstrate any statistically significant gender predisposition [14,16,19].

Studies mostly elicited the psychological and organic aspects behind sleep disturbances during COVID pandemic [9]. However, the current study also considered the different circumstantial factors like concerns over maintaining the restrictions, non-availability of the essential items, apprehensions over job situation and also infection scenario etc. These factors may be considered as stressors or may be inducers of anxiety. But in fact, these are some distinct circumstances that directly impact the psychological well-being resulting into poor sleep. While the role of anxiety in developing insomnia is important in devising long-term treatment plans, the other circumstantial factors as outlined, boast their importance in being modifiable with policy interventions. Apparently, these factors may appear short-lived, but conceptually these factors when restored to normalcy, only then the behavioral interventions on psychological determinants of sleep can have a sustained effect.

4.3. Strengths and limitations

The current study is one of the few online surveys to utilize a probability sampling technique in its design. Though the minimum required sample size was reached, but responses from several zones were suboptimal. However, by cluster adjusted robust standard errors the prevalence measures were considered valid based on the sample. The Cronbach's alpha coefficient for PSS4 was found

to be weak in the current study. This may be attributed to a reduced variability in item-specific responses, which may be due to matching levels of perceived stress among the respondents. In the regression analysis the current article considered statistical adjustment for direct effects undermining probable interactions of different variables. Generalized ordinal regression model strengthened the base of statistical inferencing. Although with even a higher power, constrained estimation of parameters with structural equations modeling can be an alternative to measure the effect sizes.

When considering the distribution of responses over the duration of data collection, majority of the responses were obtained during the early phases with a decreasing trend in new responses since the later second week of data collection period. Though it is a known fact that addiction is an important correlate of mental health, but enquiry on the same through self-reported responses were not considered in the final survey, owing to high non-response to these questions during pretesting which in turn may have been due to social desirability [6,32]. Self-reported responses are often considered biased to some extent, but response validation with retest among a random sample is helpful in terms of data integrity and validity. Apart from the design and model related issues, the very knowledge on sleep disturbance and its correlates during the lockdown phase in India is in itself a virtuous topic. Various researchers in different countries have studied some selected factors of insomnia, but the current article included various aspects contributing to sleep disturbance into a single framework and statistical model.

5. Conclusion

Proper sleep is an integral part of healthy and quality lifestyle. Considering that the study represents the national level findings among the users of different social media platforms in different age groups, the proportion of respondents reporting different levels of insomnia is not to be undermined. Suffering from any non-COVID illness, different areas of concern and apprehension related to COVID infection including the perceived severity were related to self-reported insomnia. These areas should be focused since they can be addressed through intensive tailored health communication propagated through digital or non-digital media reaching out to the members of different communities. Online/digital counseling, consisting of cognitive behavioral therapy, coping skill training, relaxation exercises and training about proper sleep hygiene techniques may be a timely solution. However, the issues of concern regarding non-availability of essentials, focus more on the demand and supply mismatch issues. These should be addressed by the governments at the national level not only by ensuring proper supply of essential items including personal protective equipment, but also through proper propagation of information to the consumers to alleviate the associated apprehension. As an established fact anxiety did have an effect on insomnia, which indicate a much-required thrust in community mental health promotion in the midst of the pandemic and especially when the country lifts the lockdown. Undoubtedly the services under National Mental Health Program in India needs to be expanded utilizing novel service delivery options as the burden observed through the lens of this online survey may only represent the tip of a dangerous iceberg.

CRediT authorship contribution statement

Arista Lahiri: Conceptualization, Data curation, Formal analysis, Methodology, Writing - original draft. **Sweety Suman Jha:** Conceptualization, Data curation, Methodology, Validation, Writing - original draft. **Rudraprasad Acharya:** Data curation, Investigation,

Validation, Writing - review & editing. **Abhijit Dey:** Data curation, Formal analysis, Investigation, Software, Visualization. **Arup Chakraborty:** Conceptualization, Data curation, Methodology, Supervision, Visualization, Writing - review & editing.

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Conflict of interest

None.

The ICMJE Uniform Disclosure Form for Potential Conflicts of Interest associated with this article can be viewed by clicking on the following link: <https://doi.org/10.1016/j.sleep.2020.11.020>.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.sleep.2020.11.020>.

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